

Unwieldy-Bench: Stress-Testing Humanoid Loco-Manipulation with Heavy, Extended, and Irregular Payloads

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Abstract: Humanoid robots are increasingly expected to operate in real-world settings that require heavy-duty whole-body transport, such as logistics, construction, and industrial material handling. However, transporting large and heavy objects fundamentally alters humanoid control dynamics: strong robot-payload coupling can induce substantial shifts in the combined center of mass, amplify angular momentum, and trigger destabilizing whole-body oscillations that challenge existing loco-manipulation controllers. We introduce Unwieldy-Bench, a benchmark for stress-testing humanoid transport with heavy, extended, and irregular payloads. Our analysis shows that representative imitation-based controllers degrade sharply in this regime, as conventional approaches primarily optimize pose and trajectory tracking while largely neglecting whole-body momentum evolution. Motivated by this observation, we introduce DDM, a lightweight dynamics-aware reward-shaping strategy that regularizes payload-induced angular momentum. Experiments demonstrate that DDM improves transportation stability, reduces object drop rates, and extends the feasible load range across diverse unwieldy transportation scenarios. Project website: <https://crossstyle.github.io/UnwieldyBench/>

Keywords: Humanoid loco-manipulation, Unwieldy object transport, Benchmarking

1 Introduction

Recent progress in humanoid loco-manipulation has been largely driven by imitation learning [11, 12, 13]. Modern humanoid controllers can reproduce diverse whole-body motions [14, 15] and support increasingly complex interaction skills, ranging from household manipulation [16, 17, 18] and dynamic athletic behaviors [19, 20, 21, 22] to object-carrying tasks [23, 24]. However, as illustrated in Fig. 1, most existing humanoid loco-manipulation benchmarks and widely used datasets [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] focus on lightweight, compact objects whose effects on whole-body dynamics are limited. In real-world applications such as logistics, construction, and industrial material handling, humanoids are expected to transport large, heavy objects that substantially affect balance, contact stability, and whole-body coordination.

Transporting such unwieldy objects places humanoids in a qualitatively different control regime, as heavy-object transport requires balance control that accounts for payload-induced shifts in the center of mass (CoM), contact stability, and whole-body dynamics [25, 26, 27]. When the payload is heavy or held with a long lever arm, the robot and object form a tightly coupled dynamical system: shifts in the combined CoM of the robot-payload system, increased rotational inertia, and amplified angular momentum can destabilize locomotion even under accurate kinematic tracking. Therefore,

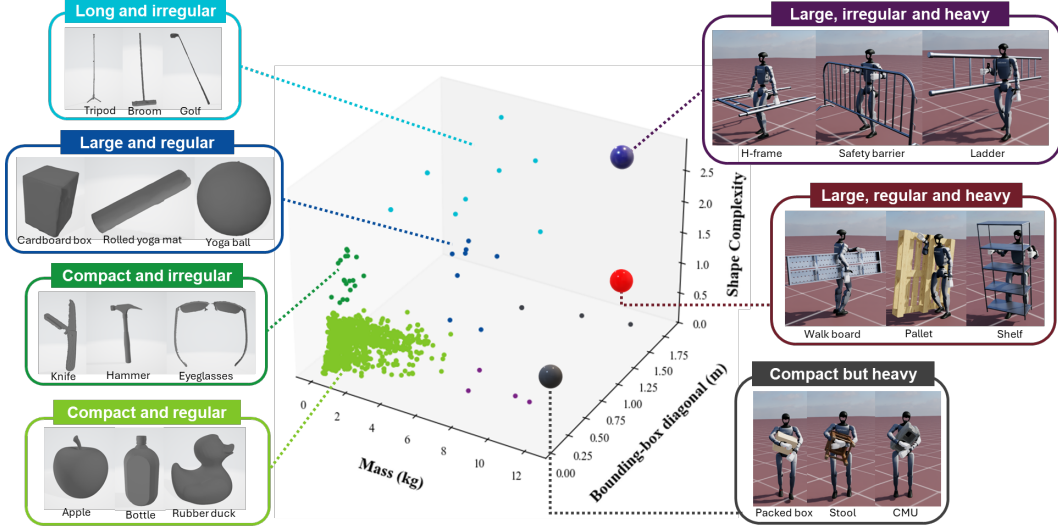


Figure 1: Object-space distribution of existing benchmarks and Unwieldy-Bench. Objects are mapped according to mass, bounding-box diagonal length, and shape complexity. Existing humanoid benchmarks and widely used datasets [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] cluster near the origin, reflecting a dominance of lightweight, compact, and regularly shaped objects. In contrast, Unwieldy-Bench spans large, heavy, and irregularly shaped payloads that induce strong robot-payload coupling and substantial whole-body balance challenges.

unwieldy object transport is not merely a more difficult instance of object carrying, but a stress test of a controller’s ability to regulate coupled robot-object dynamics beyond pure kinematic tracking.

We introduce Unwieldy-Bench, a benchmark for stress-testing humanoid transport with heavy, extended, and irregular payloads. Unwieldy-Bench includes 14 payloads spanning 5-15 kg in mass and 0.6-2.6 m in bounding-box diagonal length, instantiated across 6 carrying styles and 20 transportation scenarios. The benchmark is designed to probe failure modes that are underrepresented in existing loco-manipulation evaluations. Using Unwieldy-Bench, we evaluate representative imitation-based humanoid controllers, including DeepMimic [28], HDMI [29], AMP [30], and ADD [31], and observe severe degradation under unwieldy payloads. Notably, our analysis shows that failures often occur despite reasonable motion tracking. Heavy or spatially extended payloads can generate large angular momentum about the CoM of the coupled robot-object system. Stable transport therefore requires whole-body compensation that dissipates or redirects this momentum, rather than simply following the reference trajectory. These observations indicate that the dominant challenge lies not only in reproducing reference trajectories, but also in regulating the momentum induced by tightly coupled robot-object motion.

This finding motivates a minimal reward-shaping probe: we introduce DDM, a lightweight dynamics-aware reward-shaping strategy that regularizes payload-induced angular momentum during policy learning. It seeks to test the central hypothesis exposed by the benchmark: momentum regulation is a critical missing component for robust humanoid transport of unwieldy objects. Despite its simplicity, this shaping term substantially improves transport stability, reduces object drops, and extends the feasible load range across challenging scenarios. These results indicate that momentum regulation becomes a decisive factor when moving from lightweight object carrying to unwieldy payload transportation.

In summary, this work makes the following contributions. First, We introduce Unwieldy-Bench, a benchmark for evaluating humanoid loco-manipulation with heavy, spatially extended, and irregularly shaped payloads in transport settings. Second, we provide an empirical analysis showing that existing imitation-based controllers degrade sharply under unwieldy-object dynamics, and that these failures are often better explained by insufficient momentum regulation than by kinematic-tracking

Table 1: Comparison of existing humanoid loco-manipulation benchmarks.

Benchmark	Heavy-duty payload	Extended dimension	Irregular geometry	Carrying strategy
LocoMujoco [3]	×	×	×	Single-strategy (front-carry)
SkillBlender [1]	×	✓	×	Single-strategy (front-carry)
HumanoidBench [10]	✓	×	×	Single-strategy (torso-hugging)
Mimicking-Bench [4]	×	×	×	N/A (static manipulation)
BiGym [5]	×	×	×	Single-strategy (front-carry)
Humanoid Everyday [2]	×	×	×	Single-strategy (front-carry)
Unwieldy-Bench (Ours)	✓	✓	✓	Multi-strategy (front/side/low-carry)

error alone. Third, we show that an extremely lightweight momentum-aware reward-shaping term substantially improves stability, providing diagnostic evidence that momentum regulation is a key bottleneck.

2 Related Work

2.1 Humanoid Imitation Learning

Imitation learning has become a dominant paradigm for humanoid control, enabling policies to reproduce diverse whole-body motions from reference demonstrations. Methods based on motion imitation [28, 32, 33, 34], adversarial priors [30, 31], latent skills [35, 36], and generative policies [37, 38] have achieved strong results in locomotion [14, 15, 22] and loco-manipulation [23, 29, 39, 40, 41]. However, most imitation objectives remain centered on pose fidelity, trajectory tracking, and kinematic consistency. These objectives are effective when objects only weakly perturb the robot, but they do not directly constrain the momentum evolution of a tightly coupled robot-payload system. As shown in later experiments, under heavy or spatially extended loads, policies can track reference motions reasonably well while still becoming unstable due to payload-induced angular momentum amplification. This observation motivates evaluation that goes beyond kinematic tracking.

2.2 Humanoid Loco-Manipulation Benchmarks

Recent benchmarks such as LocoMujoco [3], Mimicking-Bench [4], SkillBlender [1], BiGym [5], HumanoidBench [10], and Humanoid Everyday [2] have expanded humanoid evaluation from locomotion to whole-body interaction, manipulation, and everyday tasks. However, their object-transport settings remain dominated by lightweight, compact, or geometrically regular objects, where object dynamics are often secondary to locomotion and end-effector tracking. Although HumanoidBench [4] and SkillBlender [1] include tasks involving heavier or larger packages, they do not systematically cover payloads that are simultaneously heavy, spatially extended, irregular, and inertially challenging under diverse carrying strategies. Unwieldy-Bench complements these benchmarks by targeting this underexplored regime, in which the object substantially affects balance, momentum evolution, and whole-body coordination. Table 1 summarizes the comparison.

2.3 Momentum Regulation in Humanoid Control

Momentum regulation is a long-standing principle in humanoid balance and whole-body control, with classical methods explicitly reasoning about CoM motion, contact forces, and centroidal angular momentum [42, 43, 44, 45]. Our work shows that this principle becomes a decisive missing factor when imitation-based humanoid controllers are evaluated under unwieldy payload dynamics. By using a minimal momentum-aware reward as a diagnostic intervention, we connect classical momentum-aware control with modern imitation-based loco-manipulation and demonstrate that failures in heavy-object transport cannot be explained by trajectory-tracking error alone.

99 3 Unwieldy-Bench

100 Unwieldy-Bench evaluates humanoid transportation in regimes where payloads substantially per-
101 turb whole-body dynamics. Instead of treating object carrying as end-effector tracking with a pas-
102 sive load, the benchmark stresses three factors that commonly arise in real-world transportation:
103 inertial coupling from heavy payloads, geometric constraints from large or irregular objects, and
104 style-dependent loading patterns caused by different payload placements, moment arms, and support
105 structures. We instantiate these factors by varying object mass, spatial extent, and shape complexity,
106 and by combining them with multiple carrying styles.

107 3.1 Unwieldy Object Taxonomy

108 We characterize unwieldy payloads along three measurable axes: mass, size, and shape complex-
109 ity. Mass quantifies the inertial load relative to the humanoid payload capacity; high-mass objects
110 increase torque demands, CoM shifts, and momentum disturbances. Size is measured by the object
111 bounding-box diagonal relative to the humanoid arm span; large objects introduce long moment
112 arms and constrain feasible whole-body postures. Shape complexity captures deviations from com-
113 pact convex geometry using a rugosity index computed from the area of object mesh A_{mesh} and its
114 convex hull A_{hull} , defined as $|1 - A_{\text{hull}}/A_{\text{mesh}}|$; irregular objects complicate stable grasping and
115 amplify orientation-dependent inertial effects.

116 For the Unitree G1 platform, we define thresholds for mass, size, and shape complexity as $M_{\text{thr}} =$
117 3 kg , $D_{\text{thr}} = 2L_{\text{arm}} = 0.90 \text{ m}$, and $C_{\text{thr}} = 1$, respectively, where L_{arm} denotes single-arm reach.
118 Based on these criteria, Unwieldy-Bench categorizes payloads into three regimes: compact heavy
119 objects, large regular heavy objects, and large irregular heavy objects. Compact heavy objects retain
120 simple geometry and exceed only the mass threshold; large regular heavy objects exceed the mass
121 and size thresholds; and large irregular heavy objects exceed all three thresholds. Full object-level
122 physical properties are reported in Appendix A.

123 3.2 Unwieldy Object Transportation Task Suite

124 Unwieldy-Bench combines each object category with multiple carrying styles, including front carry-
125 ing, torso-hugging, shoulder carrying, side carrying, low carrying, and pull-back transport (Fig. 2).
126 These styles change the location of the payload and contact structure relative to the body, inducing
127 distinct CoM shifts, moment arms, workspace constraints, and loading asymmetries. These object-
128 style combinations form transportation scenarios that range from compact heavy-object carrying to
129 large irregular-payload transport, where robot-object coupling strongly affects balance, momentum
130 evolution, and whole-body coordination.

131 3.3 Demonstration Generation

132 We generate demonstrations using a simulation-based pipeline that decouples kinematic interac-
133 tion from dynamic stabilization. A pretrained locomotion controller [46] drives lower-body motion,
134 while bimanual inverse kinematics [47] specifies upper-body object contacts and carrying poses.
135 During reference generation only, objects are assigned negligible mass to avoid inertial failures and
136 efficiently produce kinematically plausible humanoid-object trajectories. During policy training and
137 evaluation, all objects are restored to their specified mass and inertia properties. This pipeline en-
138 ables scalable generation of transport references across object categories and carrying styles without
139 manual motion capture or task-specific trajectory optimization.

140 3.4 MDP Formulation and Evaluation Protocol

141 We train each transport policy using a motion-imitation reinforcement learning formulation as a
142 Markov Decision Process (MDP), $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$. At each time step t , the state $s \in \mathcal{S}$
143 contains the proprioceptive state, the object state, and task-relevant reference information. The

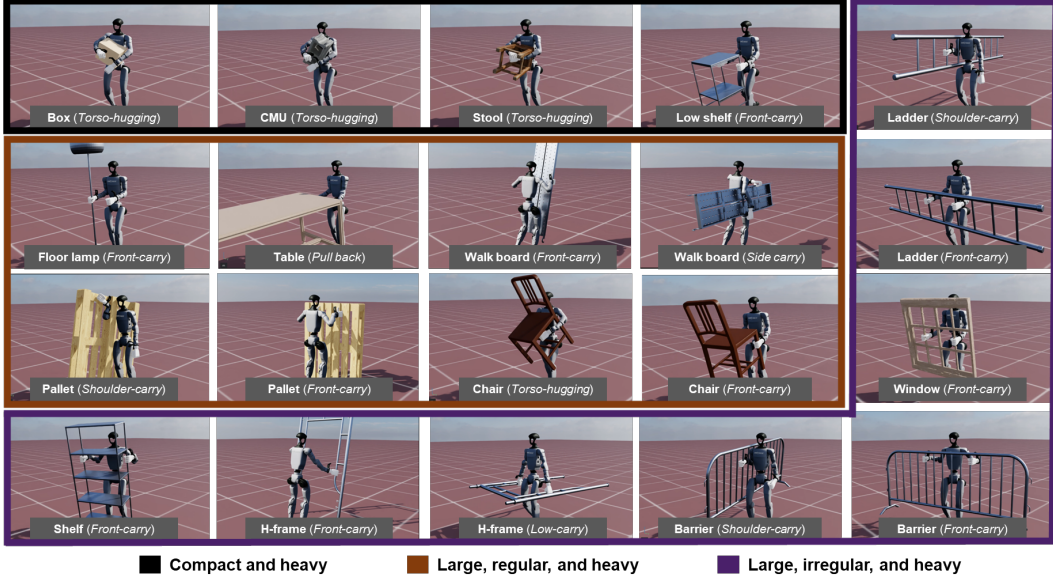


Figure 2: Unwieldy-Bench transportation task suite. Each tile shows one object-style transportation task. Border colors denote object categories: compact-heavy, large-regular-heavy, and large-irregular-heavy payloads

144 action $a \in \mathcal{A}$ specifies target joint positions, which are converted into actuation torques by low-level
 145 PD control. The transition function $\mathcal{T}$ is governed by the coupled humanoid-object dynamics in
 146 simulation. The reward $\mathcal{R}$ combines humanoid motion tracking and object pose tracking terms, and
 147 the policy is optimized to maximize the expected discounted return.

148 A trial succeeds if the humanoid transports the object beyond the target distance within the evalu-
 149 ation horizon without falling or dropping the object. Robot falls are detected when non-foot body
 150 links touch the ground or other balance-related termination conditions are triggered; object drops
 151 are detected when the object CoM falls below a predefined height threshold. We report task com-
 152 pletion metrics (success rate, time-to-termination, and transport distance), failure metrics (robot fall
 153 rate and object drop rate), and diagnostic metrics (joint-angle tracking error and payload-induced
 154 centroidal angular momentum). The last metric measures the angular momentum about the coupled
 155 humanoid-object CoM, quantifying payload-driven momentum amplification during transportation.

156 4 Experimental Results

157 4.1 Implementation Details

158 We evaluate four representative high-difficulty tasks from Unwieldy-Bench: front-carry transport
 159 with the Chair, Barrier, H-frame, and Ladder payloads. These tasks combine large spatial extent,
 160 nontrivial geometry, and pronounced payload-induced CoM offsets. We focus on front carrying
 161 because it places the payload away from the torso support region, creating larger moment arms
 162 and stronger robot-object coupling than body-supported styles such as torso-hugging or shoulder
 163 carrying. A pilot comparison across representative carry-styles in Appendix B confirms that front
 164 carrying is the most failure-prone setting, supporting its use as the primary stress-test configuration.

165 We compare representative imitation-learning baselines from two paradigms: explicit reference
 166 tracking (DeepMimic [28] and HDMI [29]) and adversarial imitation (AMP [30] and ADD [31]).
 167 For methods originally designed for locomotion or motion imitation, we adapt them to loco-
 168 manipulation by adding object pose and velocity observations and, where applicable, object pose

Table 2: Failure analysis for ladder-carrying task (15kg).

Method	Success $\uparrow$	Robot Fall $\downarrow$	Object Drop $\downarrow$	Time-to-Term (s)	Transport Dist (m) $\uparrow$	DoF Pos Err (rad)
DeepMimic	0.094 ± 0.012	0.002 ± 0.003	0.906 ± 0.012	7.043 ± 0.577	4.105 ± 0.326	0.446 ± 0.000
HDMI	0.029 ± 0.007	0.010 ± 0.006	0.965 ± 0.007	2.743 ± 0.230	1.795 ± 0.174	0.235 ± 0.001
AMP	0.039 ± 0.028	0.000 ± 0.000	0.961 ± 0.028	1.843 ± 0.186	1.600 ± 0.224	0.083 ± 0.001
ADD	0.117 ± 0.050	0.000 ± 0.000	0.883 ± 0.050	4.760 ± 0.861	2.893 ± 0.554	0.242 ± 0.001

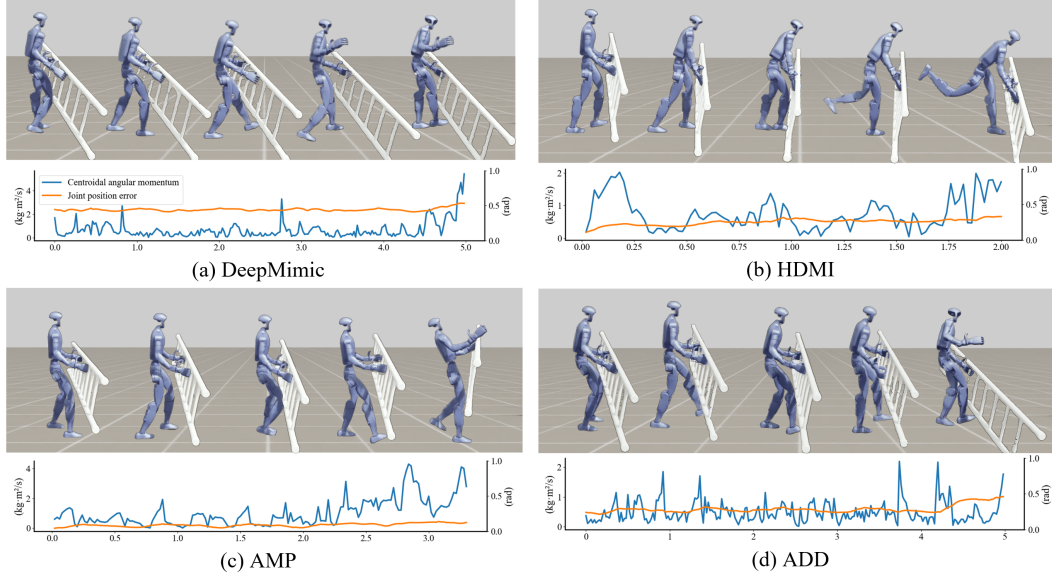


Figure 3: Qualitative results of ladder carry. Orange lines represent joint position tracking errors and blue lines show the payload-induced angular momentum (5 s before termination).

and velocity tracking rewards. All baselines use the same demonstrations, policy architecture, and evaluation protocol. Detailed implementations are provided in Appendix C.

Experiments are conducted in Isaac Sim with a Unitree G1 humanoid and two Inspire dexterous hands. To focus the benchmark on whole-body transportation rather than dexterous grasp synthesis, we use binary open-close hand commands, which reduces training complexity while preserving object support and carrying interactions. Terrains are kept flat to isolate payload-induced difficulty. Results are averaged over 4 seeds and 128 evaluation episodes per seed. Each training and testing episode lasts 10 s and 20 s, respectively. For the selected four scenarios, target transport distance is 10 m.

4.2 Failure Analysis: Tracking Is Not Enough

We analyze the ladder front-carry task to diagnose why imitation-based controllers fail under unwieldy transportation. As shown in Table 2, although robot fall rates remain near zero, object drop rates exceed 88% for all baselines. Notably, low joint-angle tracking error does not predict successful transportation: AMP attains the smallest tracking error but still drops the object in most trials and travels the shortest distance among the baselines. *This indicates that accurate kinematic imitation alone is insufficient for stabilizing the coupled humanoid-object system.*

Fig. 3 further shows that payload-induced centroidal angular momentum can grow rapidly or exhibit large spikes while joint tracking error remains comparatively smooth. These momentum transients lead to body sway, loss of object support, and eventual failure. Together, the quantitative and qualitative results suggest that unwieldy transportation is better understood as a whole-body momentum-regulation problem than as a pure trajectory-tracking problem.

4.3 Lightweight Momentum Shaping

To test whether the failures above stem from unconstrained payload momentum, we add a single momentum-shaping term to the DeepMimic reward. This variant, Dynamic-aware DeepMimic (DDM), keeps the same demonstrations, observations, policy architecture, and object-tracking rewards as DeepMimic; only the reward is changed. We compute the payload-induced angular momentum L_o of the object about the coupled robot-object CoM and express it in the robot root frame as: $\tilde{L}_o = R_{\text{root}}^\top L_o$. We then add a lightweight shaping reward that suppresses the local roll/pitch components: $R_{\text{mom}} = \exp(-\alpha \|\tilde{L}_{o,xy}\|_2^2)$, $\alpha = 0.05$. The training reward becomes $R_{\text{DDM}} = R_{\text{track}} + \lambda_m R_{\text{mom}}$, where R_{track} is the original imitation reward and λ_m controls the shaping weight. DDM is a minimal diagnostic probe for testing whether momentum regulation is a decisive bottleneck in unwieldy payload transportation.

4.4 Lightweight Momentum Shaping

Table 3 reports success rates on four front-carry tasks under increasing payload masses. At 0.1 kg, most methods achieve high success rates, confirming that the demonstrations and task definitions are feasible when robot-object coupling is weak. As mass increases to 5, 10, and 15 kg, success rates drop sharply across all baselines, especially for geometrically challenging objects such as the ladder and H-frame. This confirms that Unwieldy-Bench stresses payload-induced dynamics rather than nominal motion tracking alone.

Among the baselines, DeepMimic is the most robust in the medium- and high-load regimes, despite exhibiting larger tracking errors than several alternatives in Table 2. In contrast, methods with denser keypoint tracking (HDML) or adversarial motion-distribution constraints (AMP and ADD) often fail earlier. This suggests that strict imitation can limit the ability of a policy to deviate from the reference and discover dynamically favorable compensation behaviors under heavy payloads.

DDM consistently improves over DeepMimic at 5, 10, and 15 kg across all four tasks while using the same policy architecture and demonstrations. Averaged over all 5-15 kg settings, success increases from 35.9% with DeepMimic to 43.9% with DDM. The effect is most pronounced in the hardest cases: at 15 kg, average success increases from 19.8% to 29.5%, a 49% relative improvement. The only added signal is the lightweight momentum-shaping reward. These gains support the central finding of Unwieldy-Bench: as object transport shifts from lightweight carrying to unwieldy payload dynamics, momentum regulation becomes a critical factor for robust humanoid loco-manipulation. Additional comparisons between DeepMimic and DDM on non-front-carry tasks are reported in Appendix D, showing similar trends under low-carry, shoulder-carry, and side-carry settings.

4.5 Effect of Lightweight Momentum Shaping

We isolate the effect of the shaping term by comparing DDM with DeepMimic on the ladder front-carry task. Fig. 4 shows that DDM does not improve performance by tightening kinematic tracking; its joint-position error is slightly higher than that of DeepMimic. Instead, DDM keeps payload-induced centroidal angular momentum bounded, reduces spikes in object angular velocity, and maintains a more stable projected CoM margin. In contrast, DeepMimic exhibits increasing momentum variance and object oscillations near failure. Thus, the lightweight shaping term improves the relevant dynamics rather than imitation accuracy, supporting the view that momentum regulation is central to unwieldy object transport.

5 Limitations

This work has several limitations. First, momentum-aware shaping may bias policies toward conservative behaviors, improving stability at the possible cost of motion naturalness or transport efficiency. Second, Unwieldy-Bench focuses on the transportation phase and assumes that the object is already grasped or supported; approach, grasping, lifting, and full pick-and-place pipelines are

Table 3: Success rates on representative Unwieldy-Bench front-carry tasks under increasing payload masses.

Task	Method	0.1kg	5kg	10kg	15kg
Chair front-carry	DeepMimic	0.969 $\pm$ 0.015	0.596 $\pm$ 0.101	0.240 $\pm$ 0.053	0.018 $\pm$ 0.017
	HDMI	0.971$\pm$0.012	0.133 $\pm$ 0.016	0.000 $\pm$ 0.000	0.002 $\pm$ 0.003
	AMP	0.840 $\pm$ 0.032	0.217 $\pm$ 0.063	0.107 $\pm$ 0.017	0.115 $\pm$ 0.020
	ADD	0.945 $\pm$ 0.006	0.307 $\pm$ 0.003	0.009 $\pm$ 0.013	0.004 $\pm$ 0.007
	DDM (Ours)	0.969 $\pm$ 0.015	0.637$\pm$0.032	0.273$\pm$0.015	0.131$\pm$0.020
Barrier front-carry	DeepMimic	0.965 $\pm$ 0.004	0.728 $\pm$ 0.060	0.830 $\pm$ 0.045	0.680 $\pm$ 0.030
	HDMI	0.998$\pm$0.003	0.113 $\pm$ 0.042	0.002 $\pm$ 0.003	0.000 $\pm$ 0.000
	AMP	0.730 $\pm$ 0.048	0.444 $\pm$ 0.032	0.092 $\pm$ 0.011	0.002 $\pm$ 0.003
	ADD	0.795 $\pm$ 0.029	0.377 $\pm$ 0.046	0.541 $\pm$ 0.012	0.152 $\pm$ 0.014
	DDM (Ours)	0.965 $\pm$ 0.004	0.775$\pm$0.019	0.856$\pm$0.040	0.709$\pm$0.042
Ladder front-carry	DeepMimic	0.992$\pm$0.006	0.516 $\pm$ 0.045	0.321 $\pm$ 0.073	0.094 $\pm$ 0.012
	HDMI	0.965 $\pm$ 0.012	0.393 $\pm$ 0.040	0.125 $\pm$ 0.021	0.029 $\pm$ 0.007
	AMP	0.795 $\pm$ 0.027	0.314 $\pm$ 0.086	0.098 $\pm$ 0.047	0.039 $\pm$ 0.028
	ADD	0.977 $\pm$ 0.015	0.420 $\pm$ 0.036	0.137 $\pm$ 0.039	0.117 $\pm$ 0.050
	DDM (Ours)	0.992$\pm$0.006	0.686$\pm$0.056	0.395$\pm$0.058	0.193$\pm$0.087
H-frame front-carry	DeepMimic	0.932 $\pm$ 0.041	0.269 $\pm$ 0.072	0.020 $\pm$ 0.041	0.000 $\pm$ 0.000
	HDMI	0.986$\pm$0.008	0.080 $\pm$ 0.037	0.002 $\pm$ 0.003	0.000 $\pm$ 0.000
	AMP	0.908 $\pm$ 0.032	0.187 $\pm$ 0.058	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	ADD	0.967 $\pm$ 0.009	0.231 $\pm$ 0.023	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	DDM (Ours)	0.932 $\pm$ 0.041	0.277$\pm$0.009	0.182$\pm$0.037	0.148$\pm$0.045

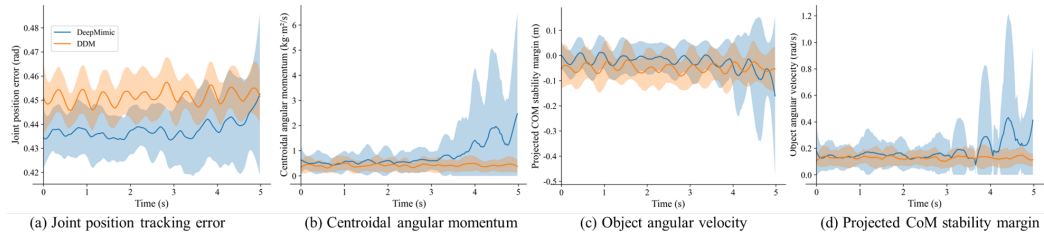


Figure 4: Effect of momentum-aware shaping on the ladder front-carry task. DDM is compared with DeepMimic. Shaded regions denote variation across trials.

left for future work. Third, the current benchmark considers rigid objects with fixed geometry, excluding deformable, articulated, or compliant payloads. Finally, all experiments are conducted in simulation. Real-world deployment will require handling contact uncertainty, sensing noise, actuation limits, and hardware safety constraints.

6 Conclusion

We presented Unwieldy-Bench, a stress-test benchmark for humanoid transportation with heavy, spatially extended, and irregular payloads. The benchmark reveals that imitation-based controllers can fail under unwieldy loads despite reasonable tracking accuracy, exposing momentum regulation as a key bottleneck beyond kinematic imitation. As a diagnostic intervention, an extremely lightweight momentum-shaping reward substantially improves stability without changing the policy architecture or demonstrations. These results suggest that momentum regulation becomes decisive when moving from lightweight object carrying to unwieldy payload transportation, and provide a benchmark for future work on physically grounded humanoid loco-manipulation.

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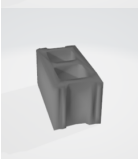
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1 A Unwieldy Object Library and Carrying Style Definitions

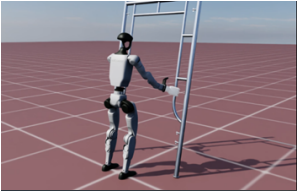
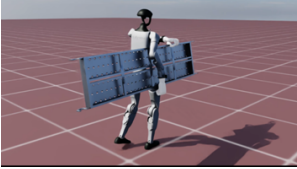
Table A1: Physical properties of the Unwieldy-Bench object library. Object size is measured by the bounding-box diagonal.

Type	Image	Name	Mass (kg)	Size (m)	C_{shape}
Compact and heavy		Stool	5–15	0.626	0.169
		Low shelf	5–15	0.826	0.272
		Box	5–15	0.423	0
		CMU	5–15	0.492	0.255
Large, regular, and heavy		Pallet	5–20	1.571	0.280
		Chair	5–15	1.206	0.633
		Shelf	5–15	1.414	0.267
		Floor lamp	5–10	1.686	0.736
		Walk board	5–20	1.726	0.319

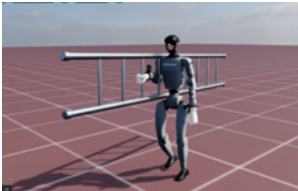
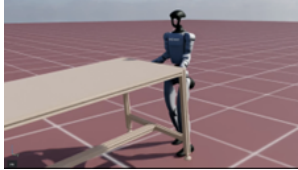
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Type	Image	Name	Mass (kg)	Size (m)	C_{shape}
		Table	5–20	1.935	0.334
Large, irregular, and heavy		H-frame	5–20	2.051	1.168
		Window frame	5–20	1.416	1.298
		Ladder	5–15	2.637	1.333
		Barrier	5–15	2.208	2.608

Table A2: Carrying styles in Unwieldy-Bench.

Carrying style	Image	Remarks
Front carry		Payload is held in front of the torso, creating a large forward moment arm and pitch instability.
Low carry		Payload is supported near the lower body, constraining leg motion and ground clearance.
Side carry		Payload is carried on one side, inducing asymmetric loading and lateral balance demands.

Continued on next page

Carrying style	Image	Remarks
Shoulder carry		Payload is supported near the shoulder, shifting load laterally and raising balance complexity.
Torso hugging		Payload is held close to the torso, reducing moment arm but limiting arm-based compensation.
Pull back		Payload is pulled behind the body, introducing backward interaction forces and delayed object response.

2 B Carrying Style Selection

3 Appendix B reports a pilot carrying-style comparison using DeepMimic. For objects with paired
 4 style variants, front carrying is consistently more failure-prone than the corresponding body-
 5 supported or lower-moment-arm alternatives. For example, H-frame front-carry drops to 26.9%
 6 success at 5 kg and 0% at 15 kg, while low-carry maintains 79.7% and 21.5% success, respectively.
 7 Similarly, Pallet and Walk board front-carry fail completely from 5 kg onward, whereas shoulder-
 8 carry and side-carry retain nonzero success even at 15 kg. The torso-hugging Stool task also remains
 9 substantially more stable under heavy loads. These results support using front-carry as the primary
 10 stress-test configuration in the main experiments, since it places the payload farther from the torso
 11 support region and induces larger destabilizing moment arms.

Table B1: Pilot comparison of carrying-style difficulty using DeepMimic.

Object	Style	0.1kg	5kg	10kg	15kg
H-frame	Front-carry	0.932 ± 0.041	0.269 ± 0.072	0.020 ± 0.041	0.000 ± 0.000
	Low-carry	0.957 ± 0.008	0.797 ± 0.013	0.466 ± 0.048	0.215 ± 0.026
Pallet	Front-carry	0.901 ± 0.052	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
	Shoulder-carry	0.940 ± 0.035	0.478 ± 0.071	0.381 ± 0.068	0.260 ± 0.136
Walk board	Front-carry	0.935 ± 0.019	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
	Side-carry	1.000 ± 0.000	0.787 ± 0.032	0.764 ± 0.031	0.625 ± 0.037
Stool	Torso hugging	0.885 ± 0.015	0.797 ± 0.045	0.520 ± 0.036	0.457 ± 0.047

12 C Baseline Implementation Details

13 We compare representative imitation-learning baselines from two paradigms: explicit reference
 14 tracking, represented by DeepMimic and HDMI, and adversarial imitation, represented by AMP
 15 and ADD. All baselines are trained with the same loco-manipulation demonstrations, the same pol-
 16 icy network backbone, and the same evaluation protocol. The only algorithm-specific components
 17 are the imitation objective, the discriminator when one is used, and the reference-action interface
 18 used by HDMI.

19 **Common setup.** Each demonstration clip contains synchronized robot and object trajectories. At
 20 each reference time, the motion library provides the robot root pose and velocity, joint rotations
 21 and joint velocities, key-body positions, object pose, object linear and angular velocity, and optional
 22 contact or grasp labels. Dynamic objects are instantiated from the object asset specified by the task,
 23 with the initial object pose taken from the demonstration/task configuration. For locomotion-only
 24 or motion-imitation methods, we adapt the observation space to loco-manipulation by adding the
 25 object state

$$o_t = [p_t - p_t^{\text{root}}, (R_t^{\text{root}})^{-1} R_t, v_t, \omega_t],$$

26 where p_t, R_t, v_t, ω_t denote object position, orientation, linear velocity, and angular velocity.

27 For a fair comparison, all methods use the same actor and critic backbone: a fully connected MLP
 28 with ReLU activations and hidden sizes (1024, 512), followed by a diagonal Gaussian action dis-
 29 tribution. The critic uses the same backbone with a scalar value head. For AMP and ADD, the
 30 discriminator uses the same MLP backbone. Policies are trained with PPO using the shared rollout
 31 and optimization settings: discount factor 0.99, GAE parameter 0.95, PPO clip ratio 0.2, 32 simu-
 32 lation steps per update, and observation/action normalization. Unless stated otherwise, evaluation
 33 uses the deterministic mean action of the Gaussian policy.

34 **DeepMimic.** DeepMimic is implemented as a reference-tracking controller. The policy observes
 35 the robot proprioceptive state, object state, phase encoding, and a short-horizon target observation
 36 sampled from the demonstration. In our implementation the target window uses future offsets of one
 37 to three simulation steps. The loco-manipulation reward augments the standard DeepMimic pose,
 38 velocity, root-pose, root-velocity, and key-body tracking terms with object-pose and object-velocity
 39 tracking:

$$r_t = w_q e^{-\alpha_q E_q} + w_{\dot{q}} e^{-\alpha_{\dot{q}} E_{\dot{q}}} + w_{root} e^{-\alpha_{root} E_{root}} + w_{vel} e^{-\alpha_{vel} E_{vel}} \\ + w_{key} e^{-\alpha_{key} E_{key}} + w_{obj} e^{-\alpha_{obj} E_{obj}},$$

40 where E_q and $E_{\dot{q}}$ are weighted joint pose and velocity errors, E_{root} tracks root position and orien-
 41 tation, E_{vel} tracks root and object linear/angular velocities, E_{key} tracks selected end-effectors and
 42 body landmarks, and E_{obj} tracks object position and orientation. Object orientation error is mea-
 43 sured with the same quaternion geodesic distance as root and body orientations. Episodes terminate
 44 early on falls, large pose deviations, object drops, or timeout, using the same termination settings as
 45 the other baselines.

46 **HDMI.** HDMI is treated as an explicit reference-tracking baseline with a residual action interface.
 47 At each step, the demonstration supplies a reference action a_t^{ref} obtained from the reference joint
 48 targets and gripper command. The policy predicts only a residual Δa_t , and the simulator receives

$$a_t = a_t^{ref} + s \Delta a_t,$$

49 with residual scale $s = 1$ unless otherwise specified. This preserves HDMI’s reference-conditioned
 50 action structure while letting the policy correct for contacts, balance, and object dynamics.

51 For loco-manipulation, the HDMI observation contains both current and future object information.
 52 The policy receives history features such as root angular velocity, projected gravity, joint-position
 53 history, and previous residual actions; command features such as future reference body positions, fu-
 54 ture joint targets, reference phase, and a_t^{ref} ; and object features such as object position and heading
 55 in the robot frame. The critic includes root-relative object pose, future object position differences,
 56 and future object orientation differences. HDMI is trained with the same reference-tracking reward
 57 family as DeepMimic, including object pose and velocity terms, plus the shared stability regulariza-
 58 tion used for object-carrying tasks.

59 **AMP.** AMP is implemented as an adversarial imitation baseline. Unlike DeepMimic and HDMI,
 60 the AMP policy does not receive future target states or a phase-conditioned tracking reward by de-
 61 fault. It receives the current robot and object state, while the discriminator receives a history of
 62 state features sampled from both the policy rollout and the demonstrations. To adapt AMP to loco-
 63 manipulation, we add object position, object orientation, object linear velocity, and object angular
 64 velocity to the discriminator observation at every history step. A discriminator sample is there-
 65 fore formed from robot root state, joint state, selected body positions, and object state over a fixed
 66 temporal window.

67 The discriminator is trained to classify demonstration samples as positive and policy or replay sam-
 68 ples as negative using binary cross entropy with logit regularization and a gradient penalty. The
 69 policy receives the standard AMP style reward:

$$r_t^{disc} = -\log(\max(1 - D(s_t), \epsilon)),$$

70 scaled by a constant factor, where $D(s_t)$ is the discriminator probability for the policy sample. No
 71 explicit object tracking term is added to AMP unless a task reward is enabled; object matching is
 72 encouraged through the discriminator input.

73 **ADD.** ADD is implemented as a differential adversarial imitation baseline. Instead of classifying
 74 absolute state observations, ADD classifies the normalized difference between the current policy
 75 state descriptor and the time-aligned demonstration descriptor:

$$\Delta d_t = d_t^{demo} - d_t^{policy}.$$

The positive discriminator input is the zero difference, corresponding to exact matching, while negative inputs are collected from policy rollouts and replayed from a discriminator buffer. The ADD reward is computed by evaluating the discriminator on the normalized difference Δd_t .

For loco-manipulation, the descriptor d_t includes both robot and object terms. The position part contains root pose, joint rotations, body positions, object position, and object orientation. The velocity part contains root velocity, root angular velocity, joint velocities, object linear velocity, and object angular velocity. This makes the differential discriminator sensitive to both object placement and object dynamics, while preserving ADD’s original time-aligned imitation objective. When a task reward is used, it is combined with the discriminator reward using the same reward-weight interface as AMP.

D Carrying Style Selection

Appendix D evaluates whether the momentum-shaping benefit extends beyond the front-carry setting. We compare DeepMimic and DDM on four non-front-carry tasks: H-frame low-carry, Pallet shoulder-carry, Walk board side-carry, and Stool torso-hugging. Across these settings, DDM consistently improves success under heavier loads. The gains are especially visible at 10-15 kg, where DDM improves H-frame low-carry from 46.6% to 56.1% and from 21.5% to 30.1%, Pallet shoulder-carry from 78.1% to 84.0% and from 76.0% to 85.2%, and Walk board side-carry from 76.4% to 82.2% and from 62.5% to 70.5%. These results indicate that the same lightweight momentum-shaping signal benefits multiple carrying styles, supporting the claim that its effect is tied to payload-induced dynamics rather than a front-carry-specific artifact.

Table D1: Success rates of DeepMimic and DDM on non-front-carry tasks.

Object	Method	5kg	10kg	15kg
H-frame low-carry	DeepMimic	0.797 ± 0.013	0.466 ± 0.048	0.215 ± 0.026
	DDM	0.842 ± 0.045	0.561 ± 0.010	0.301 ± 0.006
Pallet shoulder-carry	DeepMimic	0.778 ± 0.071	0.781 ± 0.068	0.760 ± 0.136
	DDM	0.787 ± 0.064	0.840 ± 0.017	0.852 ± 0.015
Walk board side-carry	DeepMimic	0.787 ± 0.032	0.764 ± 0.031	0.625 ± 0.037
	DDM	0.838 ± 0.030	0.822 ± 0.017	0.705 ± 0.067
Stool torso hugging	DeepMimic	0.797 ± 0.045	0.520 ± 0.036	0.457 ± 0.047
	DDM	0.854 ± 0.022	0.568 ± 0.052	0.485 ± 0.010